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From Pedals to Potentiometers: A Multi-Layered Economic Analysis of the 'Bangla Tesla'

1. Introduction: The Driver's Dilemma

I get up at 6 AM every morning and there is Rashid Bhai cleaning his green and blue electric rickshaw carefully, which is ironically called by locals in Dhaka a **Bangla Tesla**. After earning monthly 15,000 taka on a conventional human-drawn rickshaw to feed his weaving family. Nowadays, he is able to earn Tk 35,000-40,000 in exchange a month, and he does not hesitate to state that he is able to earn more, after having repaid a Tk 90,000 loan that he borrowed at a local microfinance institution (MFI) at an interest of 22 per year. His tale is symbolic of an informal economic transformation that is overwhelming Bangladesh with over 4 million of the e-rickshaws that now ply the urban roads, in what could be the largest scrum of informal electric-vehicles in the world.

I have adopted this theme because the ruling which Mr Rashid Bhai took was as near a microeconomic case study, the real McCoy, to which all those abstractions in the economics books apply, as to real life issues and questions. This essay and its special emphasis would break down his decision on rational lines and put particular emphasis on four fundamental economic principles: **Production Process, Opportunity Cost, Market Structure (Monopoly and Perfect Competition), Game Theory, Negative Externalities and Social Costs and Behavioral Economics**. Together those lenses tell a story of aspiration, failure and the diabolical, unintentional consequences that will play through an economy when self-interests are combined when they multiply, and when they multiply exponentially beyond the common good.

2. Layer 1: The Driver's Private Economic Decision

The choice that Rashid Bhai takes commences with a personal and logical analysis. He must weigh between the adjustment in the way he produces stuff and the possibility of higher salary and the extremely high risks as well as the alternative courses of action that he is unable to pursue.

2.1 Concept 1: The Production Process, A Shift from Labor to Capital

Basically, Rashid Bhai is modifying his own **production process** of making the upgrade decision. He is moving towards a production process that is nearly purely **labor based**, into one that is **capital based**.

- **Pedal Rickshaw (Labor-Intensive):** The e-rickshaw changes this equation at a fundamental level. The capital good, the vehicle itself is now the input. The outcome is a high fixed cost: the monthly payment of debt-service of approximately Tk 3,300. A job to him is a job regardless of whether he works one day or thirty days. On the other hand, his variable costs per kilometer fall off a cliff. An electricity hogger He is not a calorie burner; he is an electricity eater (~Taka 100/day), battery corrosion, etc.
- **E-Rickshaw (Capital-Intensive):** The e-rickshaw changes this equation at a fundamental level. The vehicle has become a product of capital, the input. This leaves a heavy fixed cost: monthly payment of debt-service, approximately Tk 3,300. He pays the same sum of money on a job whether he works one day or thirty. On the other hand, his variable costs per kilometer fall off a cliff. A power hog He does not use calories; he uses electricity, like Takaki 100/day), battery corrosion and wear, etc.

It is this change of variable to fixed costs that brings on the opportunity and the threat. The loan has a huge incentive of operating as many hours as feasibly possible because of its high fixed cost structure. However, once the fixed cost has been covered, the variable cost per trip is lower, which implies that every extra fare generates a full lot more profit as compared to a pedal-powered fare. He is borrowing to amass capital on which his labor can be employed to better effect - the time-tested formula of economic elevation.

2.2 Concept 2: Opportunity Cost — The High-Stakes Gamble

The Tk 130,000 that was tied up cannot be described only by the amount of money. but the value of the **next best alternative was forgotten**. This includes:

- **Forgone Investments:** Opportunity Costs- The money would have been invested in the education of his child, a highly valued, long-lasting investment, because in Bangladesh families contribute nearly two thirds of the cost of schooling or a much safer, micro-enterprise such as a food cart, which can be established at half the cost of 50,000 to 200,000 takas.
- **Financial Risk:** The debt overhang causes large risk. Variations in income from illness or repairs may make them go into default and further poverty.
- **Physical Risk:** E-rickshaws, which are often poorly manufactured, are dangerously unsound and were associated with hundreds of fatal accidents in 2025.

Although the net benefit is positive (about Tk 16,000 extra monthly pay), this analysis does not take into account the relevant market forces and social costs of operation.

Table 1: Financial Comparison of Rickshaw Types

Cost/Income Category	Pedal Rickshaw (Monthly)	E-Rickshaw ('Bangla Tesla') (Monthly)
Initial Investment	Tk 30,000	Tk 130,000
Monthly Operating Costs	Tk 1,000 (Maintenance)	Tk 4,700 (Charging & Maintenance)
Loan Payments	None	Tk 3,300 (2-year loan)
Gross Monthly Income	Tk 15,000	Tk 38,000
Net Monthly Income	Tk 14,000	Tk 30,000

3. Layer 2: The Broader Market and Social Dynamics

The personal choice of Rashid Bhai does not occur in the vacuum. It is influenced by the market structure in which he is entering and in a sense, leads to more general societal results.

3.1 Concept 3: Market Structure — From Perfect Competition to Local Monopoly

There is something puzzling about Bangladesh e-rickshaw, a mixture of rivalry and monopoly. The vehicles properly promote, on the surface, one that resembles that of **nearly perfect competition**. It is a low-tech product, there are many small informal garages distributed, thousands of production devices, and no one organization can monopolize the importation or assembly of the parts. This is the competition that also contributes in keeping the up-front purchase price quite low.

The portrait of perfect competition breaks when we consider the complementary markets in which a driver like Rashid Bhai has to walk a thin-line. In this case, he often deals with local **monopoly power**:

- **Financing:** There are formal MFIs, but a good number of drivers depend on informal credit with a local mohajon, or lender, and he might be their sole provider of capital in their extremely local area. This lender possesses monopoly power and is able to create a world in which it imposes usury interest rates far in excess of the competitive market rate and which takes much of the surplus of the driver.
- **Charging and Maintenance:** There may be only one or two garages in any particular neighborhood, with the capacity of dozens of e-rickshaws spent overnight, or of providing service by mending a special motor. They are the monopolists of this town, who charge rather more than would be the case in a hypothetical situation in which these monopolists might face the danger of being overtaken, but to which drivers have no dependable means of escaping; in this garage there is really no other way of getting charge into the plug.
- **Syndicates:** The whole unofficial rickshaw industry is a stronghold of syndicates which operate routes and unofficial "Licenses." These are cartel-type companies, where they act as a form of monopoly by throwing arbitrary barriers to entry, and by collecting rents, but this

time in the form of daily fees charged to the driver.

Thus as Rashid Bhai purchases his vehicle in an apparently competitive market, he must work within a chain of local monopoly that gradually decreases his possible profit and well-being.

3.2 Concept 4: Game Theory in Price Discrimination in the Fare Market (The Driver)

There might be opportunities where the e-rickshaw driver, in an informal manner, exercises **third-degree** price discrimination by collecting a varying effective price according to the implied willingness to pay(WTP) of passengers.

Economic Story: A driver has no freedom in advertising a variety of prices. Rather, he interprets the type of a passenger **based on his/her cues (clothing, language, urgency)**. On the fickle (e.g., business people, **tourists**) he charges a **high cash fare**, and on the more price sensitive (students, locals) charges a low cash fare.

Rashid Bhai Passenger Negotiation: A Game Theory Puzzle Part A, The game of bargaining between Rashid Bhai and his riders (A) The game is a classical game theory game, which reflects a very interesting strategy behavior: Payoffs: When the driver quotes a high fare and the passenger accepts the quote they share the fare and the game is over In this case we see the driver will maximize revenue; the passenger is maximizing the cost.

Table 2 : Fare Negotiation Payoff Matrix (Tk)

	Passenger: Accept	Passenger: Bargain
Driver: High Fare (50)	(50, -50)	(40, -40)
Driver: Low Fare (30)	(30, -30)	(25, -25)

This game has no pure strategy **Nash Equilibrium** at which the players have a best answer that is not contingent on what the other thinks. In fact, drivers are taught reputation management techniques:

- Rashid Bhai is not an exception to price discrimination; travelers are charged a higher price (guess that they can afford it) and natives are charged a lower price (they know the market).
- Familiar customers possess signalling strategies: Comment that they are local customers or they will not charge high.
- The resulting equilibrium is a high driver quote, low-price passengers, they meet at Tk 40, and it is a win-win game.

This game demonstrates how informal markets can build complex systems of self-regulation in which there are no price controls.

3.3 Concept 5: Negative Externalities and Social Costs

This transition to the e-rickshaw creates high negative externality costs to society, not accounted in the personal decision-making calculus of the driver:

Environmental Externality: Lead-acid batteries are disposed of in an inappropriate way that causes lead to escape to the soil and the water table. The battery is composed of 10-15 kg of lead and Sulfuric acid which is typically discarded in open soils or water. A part of the social cost of cleaning up the environment and the health consequences (lead poisoning) is shifted away from the driver's own private costs.

Congestion Externality: If e-rickshaw population continues to increase without any restrictions, the Optimal estimation: Traffic congestion e-rickshaws are estimated to be around 4 million; causing congestion externality in the urban streets. Every extra e-rickshaw delays all other vehicles on road, resulting in a negative network externality that the **social marginal cost > the private marginal cost**.

Safety Externality: E-rickshaws are low on safety and have higher penetration of accidents. Social costs of emergency services, health, and loss of productivity are not of an individual driver, but rather on the society.

Table 3: Private vs. Social Costs of E-Rickshaw Operation (Estimates)

Cost Type	Private Cost (Borne by Driver)	Social Cost (Borne by Society)
Vehicle Operation	Tk 4,700/month	Tk 4,700/month
Battery Disposal	Tk 0 (often dumped illegally)	Tk 2,500/month (est. health & cleanup)
Congestion	Tk 0 (driver is part of the problem)	Tk 1,800/month (est. lost productivity)
Accidents	Tk 500/month (minor repairs)	Tk 1,200/month (est. healthcare/emergency)
Total Monthly Cost	Tk 5,200	Tk 10,200

This privately costly and socially costly cheaper causes the market to produce much more e-rickshaws than is socially optimal, and the resultant **deadweight loss is enormous**- a net loss to society as a whole (and a gain to none in particular).

3.4 Concept 6: Behavioral Economics & The Irrational Decision

The math figures might appear to work in your favor, only, more closely, a closer observation of the behavioral economics will reveal why this decision rarely works into being a rational choice theory:

Optimism Bias: Drivers will not pay much attention to potential risks, such as crashes, theft and battery loss. Life of lead-acid batteries (in half the e-rickshaw fleet) is 12-18 months and replacement is Tk 15,000-20,000. But Rashid Bhais waved this away: "**God will guard me**". In informal economies, where experience-based learning is more important than statistical reasoning, this optimism bias is common.

Social Proof & Herding Behavior: The rapid uptake (200,000 in 2016 to 4 million in 2025) drives network effects as drivers model themselves on their peers in the absence of an intuitive approach to working it out. Rashid Bhai explained: **"Everybody is doing it, how can I be left behind?"** This herding effect drives adoption higher, but not paying attention to an individual's financial situation.

Mental Accounting: Rashid Bhai treats the loan separately from daily earnings, creating a psychological separation between debt and income. He focuses on daily net cash (Tk 1,200-1,500) while ignoring the long-term interest burden—a classic mental accounting error.

These behavioral factors generate a gap between perceived and actual utility for a potentially suboptimal decision to spread in an informal market.

4. Graphical Analysis and Data Visualization

4.1 Graph 1: Opportunity Cost Over Time

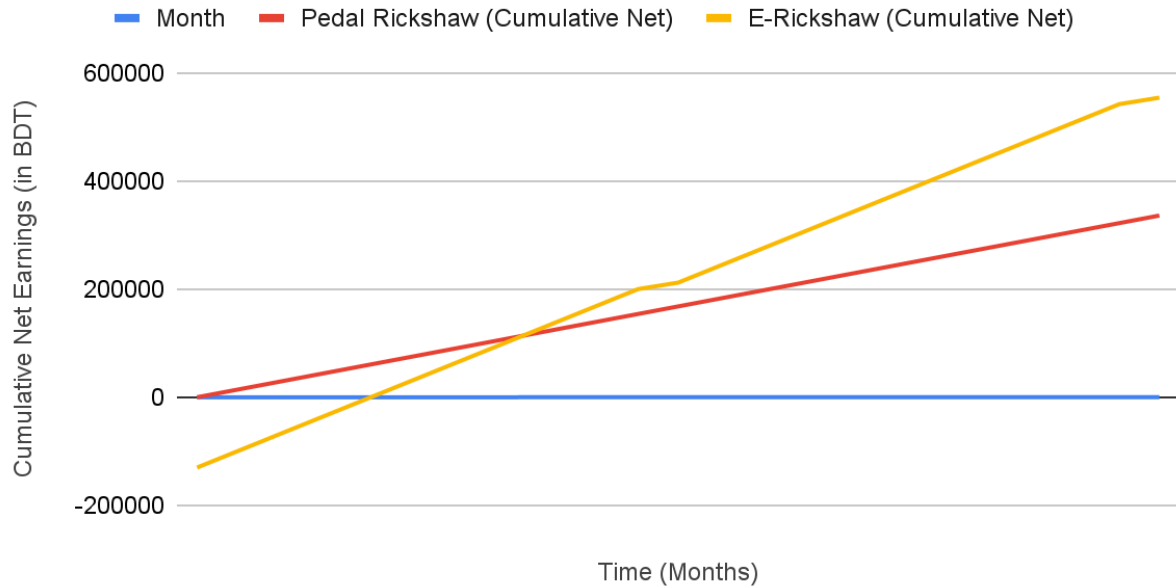
This chart illustrates the cumulative net earnings over 24 months, highlighting the initial debt of the e-rickshaw, the break-even point, and the significant recurring cost of battery replacement.

Table 4: Data for Cumulative Net Earnings Chart (Pedal vs. E-Rickshaw)

Month	Pedal Rickshaw (Cumulative Net)	E-Rickshaw (Cumulative Net)	Notes
0	0	-130000	Initial Investment
1	14000	-100000	
2	28000	-70000	

3	42000	-40000	
4	56000	-10000	
5	70000	20000	Break-Even Point
6	84000	50000	
7	98000	80000	
8	112000	110000	
9	126000	140000	
10	140000	170000	
11	154000	200000	
12	168000	212000	Battery Replacement (-18,000)
13	182000	242000	
14	196000	272000	
15	210000	302000	
16	224000	332000	
17	238000	362000	
18	252000	392000	
19	266000	422000	
20	280000	452000	
21	294000	482000	
22	308000	512000	
23	322000	542000	
24	336000	554000	Battery Replacement (-18,000)

Cumulative Net Earnings: Pedal Rickshaw vs. E-Rickshaw (24-Month Projection)

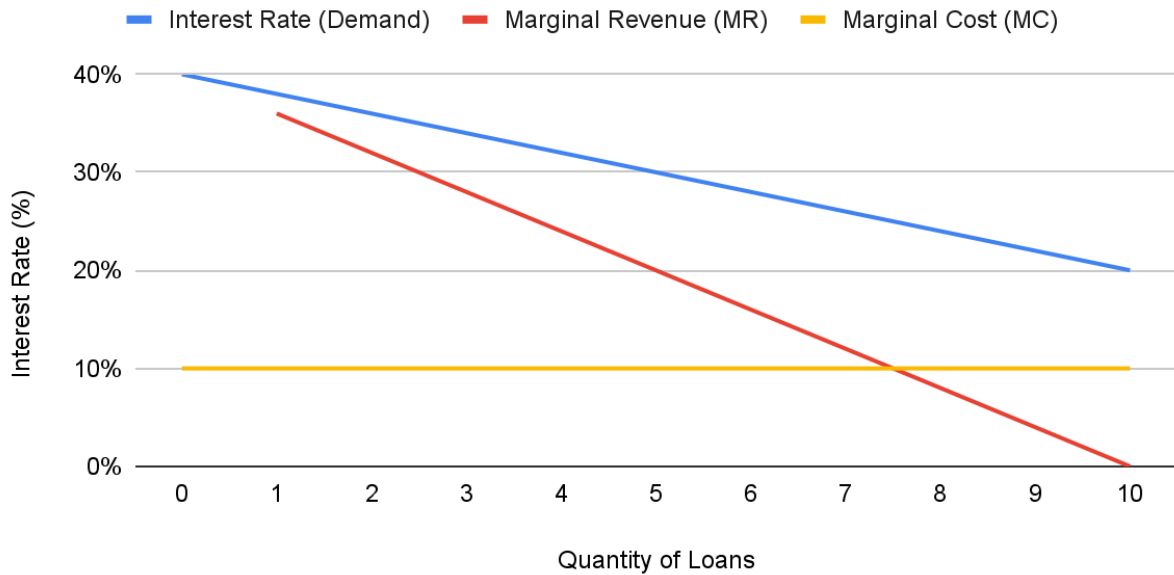


4.2 Graph 2: Market Structure & Monopoly Power (Loan Financing)

Quantity of Loans	Interest Rate (Demand)	Marginal Revenue (MR)	Marginal Cost (MC)
0	40%	-	10%
1	38%	36%	10%
2	36%	32%	10%
3	34%	28%	10%
4	32%	24%	10%
5	30%	20%	10%
6	28%	16%	10%
7	26%	12%	10%
8	24%	8%	10%
9	22%	4%	10%
10	20%	0%	10%

Monopoly Power of the Local Moneylender

How a loan shark maximizes profit by restricting credit and



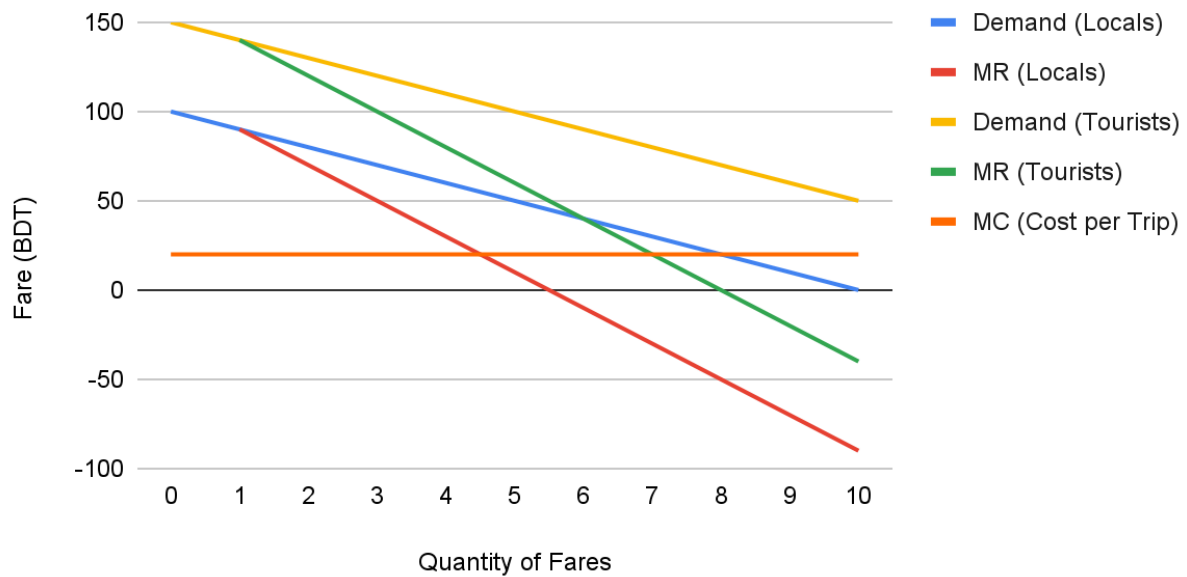
- **Monopoly Profit-Maximizing Point:** The moneylender will issue loans where $MR = MC$. In the table, this happens closest at 7 loans at a 26% interest rate.
- **Competitive Market Outcome:** In a competitive market, the rate would be set where Demand = MC (10%). At a 10% rate, drivers would demand 10 loans.

4.3 Graph 3: Price Discrimination in the Fare Market (The Driver)

Quantity (Fares)	Demand (Locals)	MR (Locals)	Demand (Tourists)	MR (Tourists)	MC (Cost per Trip)
0	100	-	150	-	20
1	90	90	140	140	20
2	80	70	130	120	20
3	70	50	120	100	20
4	60	30	110	80	20
5	50	10	100	60	20
6	40	-10	90	40	20
7	30	-30	80	20	20
8	20	-50	70	0	20
9	10	-70	60	-20	20
10	0	-90	50	-40	20

Informal Price Discrimination by an E-Rickshaw Driver

Charging different fares to tourists and locals to maximize total profit



Here,

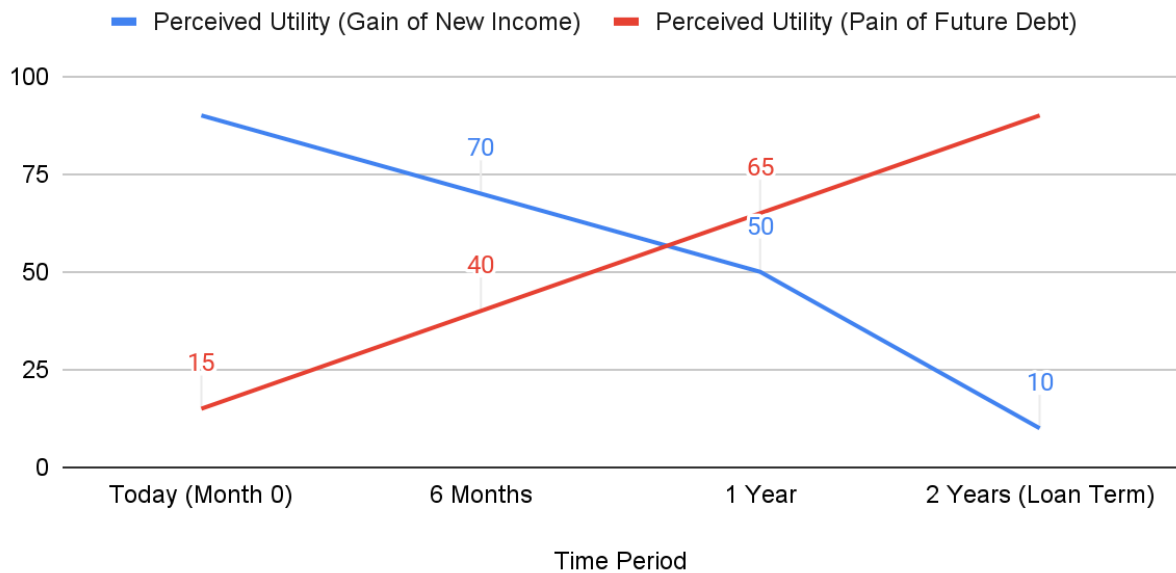
- Rashid Bhai will serve each group until **MR = MC (20)**.
- For Tourists: MR=MC at ~**6 fares**. The price tourists pay is found on their demand curve at that quantity: **90 BDT**.
- For Locals: MR=MC at ~4 fares. The price locals pay is **60 BDT**.
- **Total Profit: (Tourist Profit) + (Local Profit) = [(90-20)*6] + [(60-20)*4] = 420 + 160 = 580 BDT**. This is higher than if he charged one price!

4.4 Graph 4: Behavioral Economics (Hyperbolic Discounting)

This chart visualizes the time-inconsistent preferences that lead to a potentially suboptimal decision, like taking a high-interest loan.

Time Period	Perceived Utility (Gain of New Income)	Perceived Utility (Pain of Future Debt)
Today (Month 0)	90	15
6 Months	70	40
1 Year	50	65
2 Years (Loan Term)	10	90

Perceived Utility (Gain of New Income) and Perceived Utility (Pain of Future Debt)



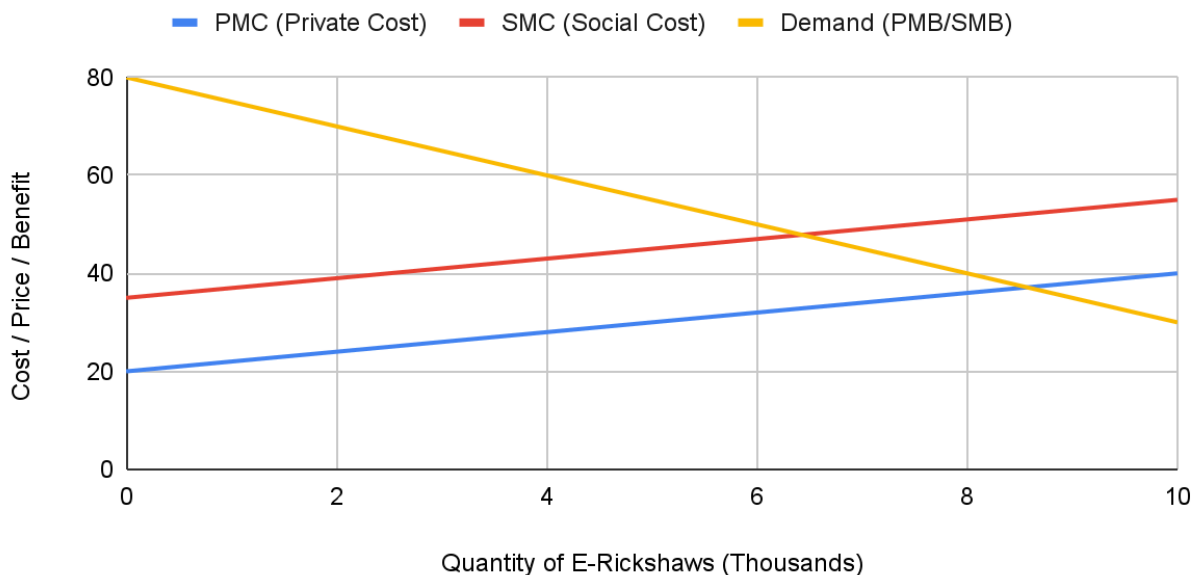
4.5 Graph 5: The Negative Externality of E-Rickshaws

Quantity of E-Rickshaws (Thousands)	PMC (Private Cost)	SMC (Social Cost)	Demand (PMB/SMB)
0	20	35	80
1	22	37	75
2	24	39	70
3	26	41	65
4	28	43	60
5	30	45	55
6	32	47	50

7	34	49	45
8	36	51	40
9	38	53	35
10	40	55	30

Negative Externality of E-Rickshaws

The Social Cost of Pollution and Congestion



This diagram illustrates the market failure caused by unpriced social costs.

- **PMC (Private Marginal Cost):** The direct costs borne by Rashid Bhai (loan, electricity, maintenance).
- **SMC (Social Marginal Cost):** The full cost to society (MPC + external costs of pollution, congestion, and accidents).
- **Market Equilibrium (Qm):** The actual number of e-rickshaws on the road, determined by private costs and benefits (Not Shown in the graph due to the complexity).
- **Socially Optimal Equilibrium (Qs):** The efficient number of e-rickshaws if all social costs were factored in (Not Shown in the graph due to the complexity).
- **Deadweight Loss:** The net loss of social welfare because the market produces too many e-rickshaws ($Q_m > Q_s$) (Not Shown in the graph due to the complexity).

5. Conclusion: Personal and Economic Insight

I perceived the e-rickshaw owned by Rashid Bhai as merely an upgrade of this vehicle, as a sign of development, before I studied economics. I now view a complicated microeconomic system. It is a rational alteration in his **production process** of labor to capital but with a high **opportunity cost**. He gets into a market that seems competitive but that is hampered by local **monopolies** that capture values. Lastly, his decision, which is individually rational, leads to a colossal **market failure** and a fall in the total **social welfare** in the form of un-priced externalities such as pollution and congestion.

The Bangla Tesla is not a vehicle but is a phenomenon of resistance to human ambition in a web of failure of institutions. Such complexities can be ascertained with the help of means that economics provides, but an ingenious policy should solve these problems in order to ensure that the person who succeeds to emerge out of poverty does not put so many people in the same situation.

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